Poemorama: A 360-Degree Poetry Guessing Game Powered by Closed-Source Image Models and Panorama Adaptation

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Abstract

Poemorama is a 360-degree guessing game for Chinese classical poetry: a player explores an AI-generated panoramic scene and guesses the poem line represented by the scene. The core product contribution is the game itself: a culturally familiar yet challenging interaction that turns poem recall into an immersive visual guessing task. On the generation side, our goal is to generate panoramas that faithfully represent poetic imagery while satisfying the geometric and perceptual requirements of 360-degree viewing. We adapt a closed-source image model (GPT-Image-2[1]) to panorama generation through panorama-specific prompt design, seam repair formulated as shifted inpainting, super-resolution workflows and experiments, and explicit geometric constraints for equirectangular imagery. Independently, we carefully design Poemorama as a game that balances playability and challenge through collection choice, line filtering, free-form guessing, Wordle-like feedback, offline data preparation, and frontend interaction. The current system is a complete and scalable product prototype with generated panorama questions, prepared data artifacts, and a playable browser frontend. The demo website is available at https://imzzy1201.github.io/poemorama/.

1 Introduction

Poemorama is a game in which players view a 360-degree poetic scene and guess the corresponding Chinese classical poem line. It is inspired by WenWare,[2] a web-based guessing game built around AI-generated 360-degree historical scenes. While WenWare combines historical events with place-and-year guessing, Poemorama combines classical poetry with Wordle-like poem-line guessing, making it more tailored to Chinese audiences. The interaction is multimodal because the system must transform compact classical Chinese language into immersive visual content whose scene, atmosphere, and concrete objects can support human recall.

[1] https://openai.com/index/introducing-chatgpt-images-2-0/
[2] https://wen-ware.com/

Preprint.

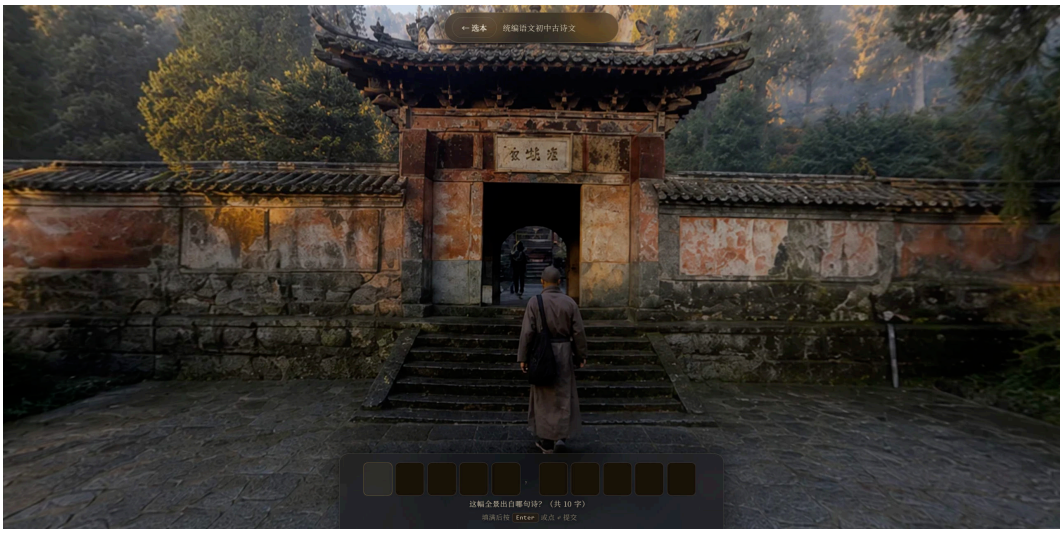


Figure 1: Overview of Poemorama: generated 360-degree poetry scenes are used as visual clues for free-form poem-line guessing with textual feedback. The answer of this example is "清晨入古寺，初日照高林".

The central challenge is generating panoramas that are both semantically expressive and suitable for spherical viewing. WenWare does not disclose technical details about how its panoramas are generated, and its public-facing panoramas still show limitations such as visible seams, insufficient resolution, and geometric distortion. Preliminary experiments suggested a tension between semantic quality and panorama validity. Open-source panorama models such as DiT360 [Feng et al., 2026] target the correct image format but often fail to capture culturally specific poetic details. Strong closed-source image models, such as GPT-Image-2, have better world knowledge and visual quality, but their outputs are not automatically valid 360-degree panoramas. This leads to the core technical question of our project: how can we use strong closed-source image models while still satisfying the special requirements of 360-degree panoramic viewing?

The report is organized around two parts. First, to answer this question, we use GPT-Image-2 with carefully tuned prompts to generate a basic WenWare-level panorama from a poem line, then refine the generated image with seam repair, super-resolution, and geometry constraints for boundary continuity, fullscreen perceptual detail, and spherical viewing validity. Second, we describe how we designed Poemorama as a game to balance playability and challenge through collection scope, data preparation, visualizability filtering, answer feedback, and frontend implementation.

2 Preliminary Experiments

We first tested whether existing image generation models could produce usable poetic panoramas from a tuned prompt (Figure 2). DiT360, an open-source panorama model, produced an image in the correct broad format but failed to capture several explicitly prompted elements, including plum blossoms, an ancient-style travel robe, and rain-soaked outer clothing. This confirmed that panorama-specific model architecture alone was not enough for culturally detailed poem scenes.

We then tested two strong closed-source image models, Nano-Banana-2[3] and GPT-Image-2, with the same tuned prompt. Both followed the semantic details much better than DiT360, producing more realistic and visually rich scenes for the same poem line. Their results are close, so we choose GPT-Image-2 for the following workflow, mainly to keep the generation pipeline consistent with our earlier experiments and proposal.

[3] https://blog.google/innovation-and-ai/technology/ai/nano-banana-2/



Figure 2: Preliminary experiments with different image generation models. Top: DiT360. Bottom left: GPT-Image-2. Bottom right: Nano-Banana-2.

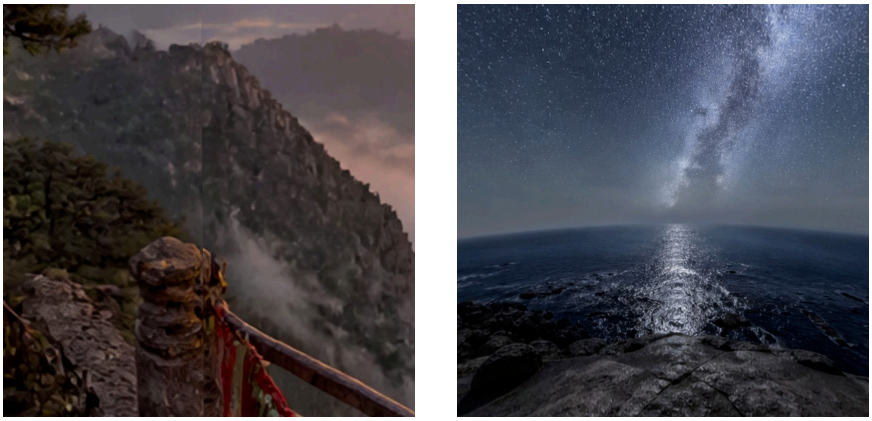


Figure 3: Visible seam (left) and geometric distortion (right) screenshots from the 3D viewer.

However, direct outputs from closed-source image models still show panorama-specific problems (Figure 3). In a 3D viewer, the left-right boundary could become a visible seam, and the projected image could show geometric distortion such as a curved horizon. These failures are not always obvious in a flat 2:1 preview, but they matter in the actual spherical viewing experience. This motivated the later seam repair, super-resolution, and geometry-constraint work.

3 Adapting Closed-Source Image Models to Panoramas

3.1 Panorama-Specific Prompting

A common way to use a text-to-image model for batch scene generation task is to first ask a language model to generate a scene prompt, then insert that prompt into a fixed task template.

We follow this general pattern, but found that both parts require careful tuning to solve the unconventional challenges of text-to-panorama generation. The first prompt must tackle the complex spatial arrangement of visual elements, ensuring they are appropriately distributed in correct positions across the 360-degree panorama. The second prompt must then force a



Figure 4: A comparison of super-resolution results: the original image (top) and the SUPIR output featuring visible face distortion (bottom).

general-purpose image model to adhere strictly to the geometric and perceptual constraints of a spherical viewer. Rather than being mere decorative instructions, this deliberate prompt tuning acts as the crucial first adapter, transforming an off-the-shelf high-quality image model into a viable engine for immersive 360-degree content.

3.2 Seam Fix as Shifted Inpainting

The most visible failure of direct panorama generation is the seam at the left-right boundary. In a flat image, the left and right edges appear far apart, but in a spherical viewer they are adjacent directions. If a model generates unrelated content at the two edges, the seam becomes immediately visible during exploration.

We convert this boundary problem into a simpler inpainting problem. Given a generated panorama, we horizontally shift it by 180 degrees. The original left-right seam is moved to the center of the image, while the original center is split across the new left and right edges. We then mask a narrow central band and ask GPT-Image-2 to fill it while minimizing changes to the rest of the image. This inpainting task is unusually well constrained: the



Figure 5: An example before (top) and after (bottom) seam repair and super-resolution; both images are horizontally shifted to position the left-right seam in the center.

region is regular, the area is small, the surrounding context is available, and the model does not need to invent a new object or scene.

However, image editing models may modify more than the masked region. After repairing the shifted center, the two far edges of the shifted image may no longer match perfectly, even though they came from a continuous region in the original image. We therefore blend the repaired shifted image with the original shifted image using horizontal weights. The repaired image contributes most near the center seam, while the original shifted image contributes most near the far left and right edges. Finally, we shift the result back. This produces a panorama that benefits from both images: one is continuous in the middle, while the other is continuous at the two ends. This results in a seamless panorama that is visually consistent across the left-right boundary.

3.3 Super-Resolution for Panoramic Viewing

Super-resolution is important because Poemorama images are not viewed as small thumbnails. A panorama can look acceptable as a flat preview but insufficient inside a 360-degree viewer.

We first explored dedicated state-of-the-art super-resolution models and upscalers, including SUPIR [Yu et al., 2024]. In practice, they introduced two problems for our use case. Local structures such as faces could degrade or become unnatural, as shown in the first row of Figure 4, and inference could be very slow, making the workflow costly and difficult to scale.

We found that using GPT-Image-2 to upscale the images resolves these issues and achieves excellent results. However, this presents a paradox: we need GPT-Image-2 to output a larger super-resolved image, yet our initially generated panoramas have already reached the model's maximum size limit. To overcome this restriction, we divide the image into four overlapping tiles, upscale each tile individually, and stitch them back together. During this process, we apply robust geometry correction (such as affine alignment and clipped optical flow) alongside global color alignment to ensure seamless integration and preserve the overall consistency of the panorama.

The second row of Figure 5 shows an example where seam repair and super-resolution together improve the final panorama.

3.4 Explicit Geometry Constraints

Beyond the basic panorama format requirements, we incorporate manually designed prompts to guide the model in generating geometrically correct panoramas. For example, we explicitly require that the horizon remains perfectly straight and is positioned exactly at the vertical center of the image, or that any two objects symmetrically opposite the camera must be separated by a horizontal distance equal to exactly half the panorama's width. We found that explicitly enforcing simple hard constraints like these in the prompt is already highly effective at preventing the most obvious geometric failures.

4 Designing Poemorama for Playability

4.1 Core Gameplay and Wordle-Style Interaction

The core gameplay of Poemorama is deliberately centered on guessing the specific poem line rather than the author, title, dynasty, or geographical place, as the individual poem line remains the definitive unit where visual imagery and textual memory naturally intersect.

To preserve this experience, the system utilizes a free-text input field instead of a multiple-choice format. A multiple-choice structure would inadvertently reduce the task to a simple process of elimination or basic visual matching. In contrast, using a free-text input is significantly more challenging, ensuring the game truly tests a player's genuine recall and memory.

To enhance playability and transform the guessing process into a progressive challenge, we adopted a Wordle-style interaction mechanism. Under this system, players receive real-time feedback mapped across individual character identities and specific pinyin components, including initials, finals, and tones. This allows them to use phonetic and structural clues to adjust subsequent guesses, thereby providing highly informative, incremental hints that guide struggling players to solve the puzzle step-by-step and enjoy the rewarding experience of gradual problem-solving.

4.2 Data Curation and Filtering Pipeline

To make the game approachable, the underlying data model decouples canonical poems from their specific collection occurrences. This separation allows players to select a familiar educational scope, such as Tang Poems 300 or junior high school textbooks, ensuring the difficulty feels manageable rather than overwhelmingly vast. We obtain canonical poem source data and the Tang Poems 300 collection information from the `chinese-poetry`[4] repository, and construct the junior and senior textbook collections with custom crawlers.

To eliminate poem lines that rely on abstract concepts, historical allusions, or non-visual descriptions, we implement an offline pipeline to select lines that depict a single, distinct

[4] `https://github.com/chinese-poetry/chinese-poetry/`

scene suitable for stable visualization. Within this pipeline, an LLM extracts candidate lines, generates interpretations, and scores them specifically on whether a line can yield a stable, unambiguous visual clue. Across our collections, roughly one-quarter of the evaluated poem lines met this threshold and were preserved for the final question bank.

Inspired by the popular Chinese Wordle game `handle`[5], we implemented a complete frontend and backend architecture tailored to this interaction model. To ensure high availability and ease of access for players, the entire application has been successfully built as a static site and deployed directly on GitHub Pages.

5 Discussion and Limitations

Poemorama is not only a text-to-image demonstration, but a complete attempt to turn generated panoramas into a playable poetry guessing game. This changes the evaluation target: an image must not only look beautiful, but also work as a 360-degree environment and as a fair visual clue for a poem line. An good image should contain recognizable poetic imagery while remaining coherent under spherical viewing.

Our main finding is that strong closed-source image models are useful for this task because they understand Chinese poetic imagery and produce high-quality scenes, but they are not directly reliable panorama generators. Raw outputs may contain visible seams, insufficient fullscreen detail, or geometric distortion. Therefore, the system adapts the model through panorama-specific prompting, shifted-inpainting seam repair, tiled super-resolution and explicit geometry constraints. These steps are not separate tricks, but a practical pipeline for converting a general-purpose image model into a usable engine for immersive game content.

The current system still has limitations. We have not yet conducted a formal user study to measure recognition rate, difficulty, or player experience. The pipeline also depends on closed-source model behavior, which limits reproducibility and makes large-scale expansion costly. Finally, only the junior textbook collection has completed the full image-side pipeline, while the other collections remain partially prepared given the cost consideration.

6 Conclusion

We present Poemorama, a 360-degree guessing game for Chinese classical poetry. We design and implement a sophisticated pipeline that adapts a strong closed-source image model to panorama generation through prompt tuning, seam repair, super-resolution, and geometry constraints. With the help of manual review and careful game design, we create a complete product prototype that turns poem recall into an immersive visual guessing experience. Our work provides a case study of how to leverage powerful multi-modal models for unconventional content generation tasks, highlighting the potential and challenges of using such models in creative applications.

Author Contributions

Chengwei Liang was mainly responsible for the game design, data curation, and the panorama generation pipeline. Zeyang Zhang was mainly responsible for the frontend development and deployment. Both authors also assisted each other throughout the project. They collaborated on the writing of this report.

[5] `https://github.com/Antfu/handle`

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